

LAWSOME

SYNTHETIC LEGAL DATA GENERATION

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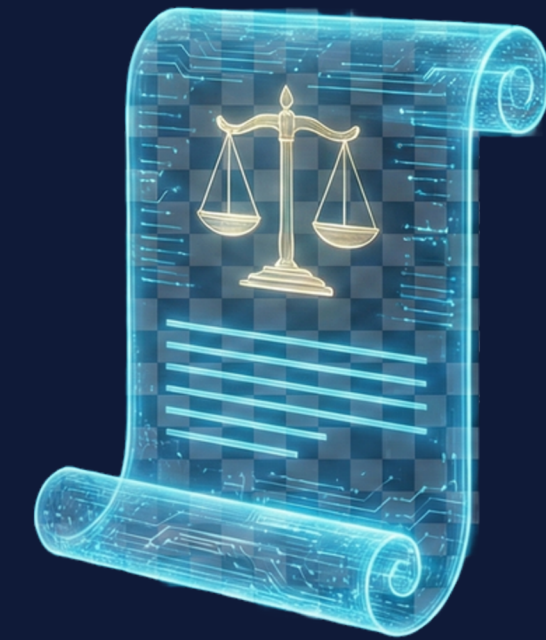
Muntean Larisa-Maria

Salejan Maria

Agoston Nikolett



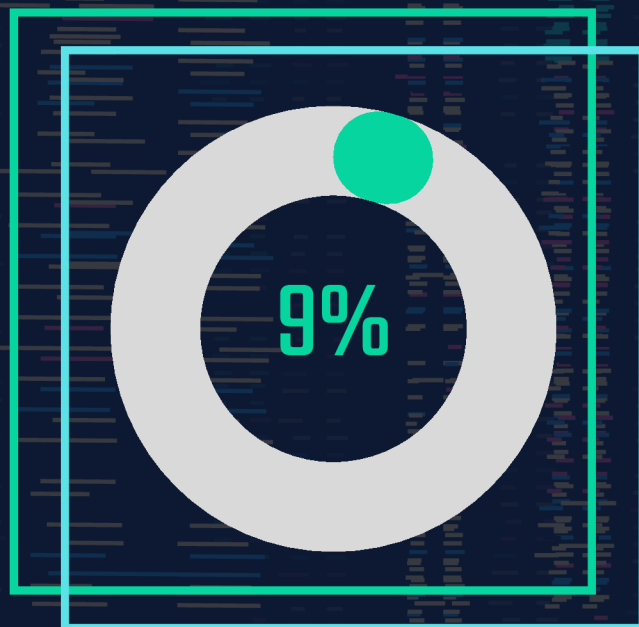
PROBLEM STATEMENT



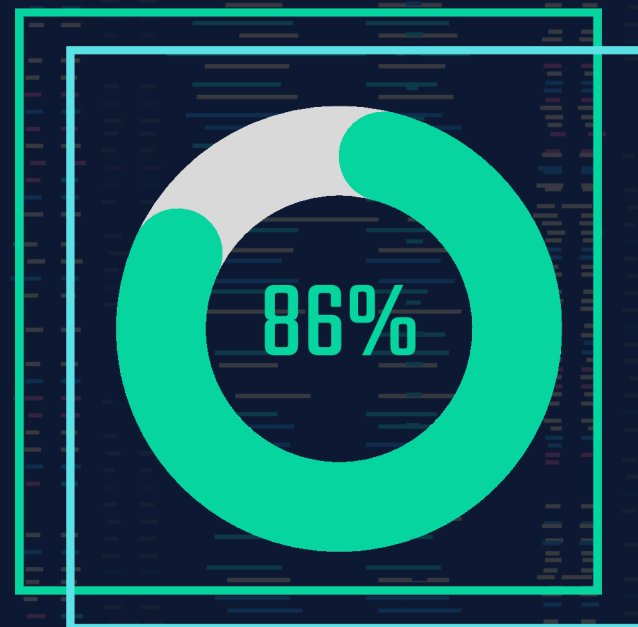
- **Synthetic legal document generation under resource constraints**
- **The foundation:** Romanian is an underrepresented language in NLP, with qualitative legal data particularly scarce.
- **Our objective:** Create a robust pipeline for generating synthetic legal data to train models on specialized legal tasks
- **Technical Constraints:** Fine-tuning LLMs is computationally expensive and resource-intensive. Legal documents demand zero tolerance for structural or terminological errors.

MARKET RESEARCH

The Opportunity Between “Free” and “Safe”



Romanian law professionals



Private law firms

Free Templates (Unsafe)



Lawsome



Lawyers (Expensive)



BUSINESS MODEL & PRICING

Flexibility and Scalability



FREE

30 starting credits
10 questions / day



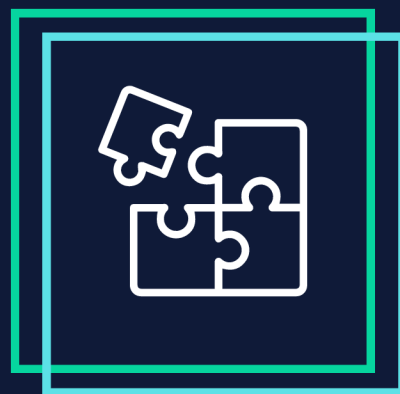
STAR

9.99 RON / month
50 credits



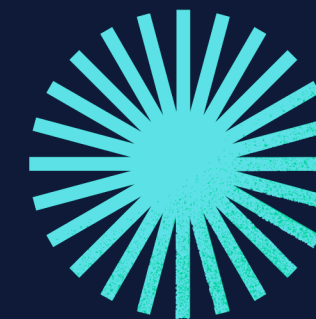
PRO

34.99 RON /month
800 credits



UNLIMITED

129.99 RON / month



WORKFLOW



REAL DATA
ACQUISITION
AND CLEANING



CREATING A
REPRESENTATIONAL
SCHEMA FOR EACH
DOCUMENT TYPE



CONTEXT-AWARE
GENERATION



EVALUATION

REAL DATA AQUISITION & CLEANING



- Webscrapping from legal sources, such as: notari.pro, legeaz.ro and portal.just.ro.
- In total, the sources provide 300+ documents, that contain placeholders (e.g.: '.....')
- We selected 3 types of legal documents to clean and synthetically generate: **requests, contracts and powers of attorney.**
- **RoLlama-8b-Instruct** was instructed to remove the placeholders and complete it with fake data and generate metadata about each document (type, purpose, short description, parties involved).
- The cleaned version of the document, along with its metadata was turned into a JSON.

GENERATING SYNTHETIC DATA

STAGE 1

- Dedicated LLM instance of RoLlama3-8b-Instruct-DPO acts as "Screenwriter"
- Generates complex JSON metadata with diverse legal scenarios
- Introduces semantic variety and realistic edge cases

STAGE 2

- Semantic matching retrieves optimal template from reference repository
- One-Shot learning paradigm
- LLM rewrites template using new synthetic metadata

THE CRITIC

- **Evaluation architecture:** Independent validation layer using RoMistral-7B-Instruct configured as specialized notarial auditor
- **Three-Dimensional Assessment Framework:**
 1. **Structural Integrity (1-10 Scale)**
 2. **Linguistic Formality (1-10 Scale)**
 3. **Purpose Alignment (1-10 Scale)**
- **Evaluation Output Format:** Structured JSON schema containing three numerical scores, plus qualitative observations field for context-specific remarks and improvement recommendations.

RESULTS



Powers of attorney

- Average **structure** score: **6.94**
- Average **language** score: **7.94**
- Average **purpose compliance** score: **6.63**



Contracts

- Average **structure** score: **8.87**
- Average **language** score: **8.37**
- Average **purpose compliance** score: **7.50**



Requests

- Average **structure** score: **7.53**
- Average **language** score: **9.00**
- Average **purpose compliance** score: **7.21**

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STRENGTHS

- No fine-tuning required
- Scalable pipeline with modular architecture
- 100% adherence to formatting through template retrieval
- Automated evaluation system

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WEAKNESSES

- No quantitative diversity metrics
- Evaluation may miss key legal quality aspects

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OPPORTUNITIES

- Expand to more legal document categories
- Integrate human-in-the-loop feedback for refinement
- Commercial applications in legal tech sector

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THREATS

- Regulatory requirements for AI-generated legal documents
- Legal liability concerns
- Need for ongoing validation by legal experts



THANK YOU!

